



Preliminary Design Review Falali - One AMR Fits All

Group 10

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Concept Summary

What are we doing?

We are developing an Autonomous Mobile Robot (AMR) to automate trolley transport within a textile manufacturing facility. Currently, workers manually push these carts between stations, a process we aim to streamline. The robot docks underneath the trolley, deploys its side-gripping arms to secure the load, and autonomously transports it to its designated destination.



Why?

- To reduce the cost of hiring manual labour just to push the trolleys
- Increase efficiency when transporting the trolleys
- To reduce the chances of collision from the trolleys and lower the chances of damaging it
- Maintenance costing RM20,000/month

Who?

- The company - lower labour cost, fewer damaged trolleys, faster transport cycles
- The company's clients - more reliable, on-time fabric handling

State of the Art



FLEXLEY MOVER® P603 PLATFORM AMR

- smaller surface area
- cannot fit under the trolley

SEAGRID TOW TRACTOR S7 AMR

- powerful but bulky
- cannot navigate tight spaces



Shortages from the current solutions in the market:

- Cannot fit under the trolley
- Require a large turning radius if coming from the sides
- No grip to carry the load
- Need to change the existing trolley to fit the AMR

Innovation In Our Project

- Low-profile under-entry capture as it drives beneath the trolley instead of docking from the side
- Auto-orientates to the trolley centre before capture
- Four-sided autonomous clamping adapts to different trolley sizes, unlike a fixed clamp
- Uses the trolley's own caster wheels and the AMR never lifts the trolley

Reference	Useful capability	Gap for this project
Platform AMR	Compact autonomous base	Top platform does not fit under trolley
Tow tractor AMR	High pulling power	Bulky and needs turning space
Side docking systems	Can engage carts	Requires trolley-side access or trolley modification
Group 10 Direction	Low-profile under-entry capture	Needs reliable hook/arm integration

AI Tools and Design Vision

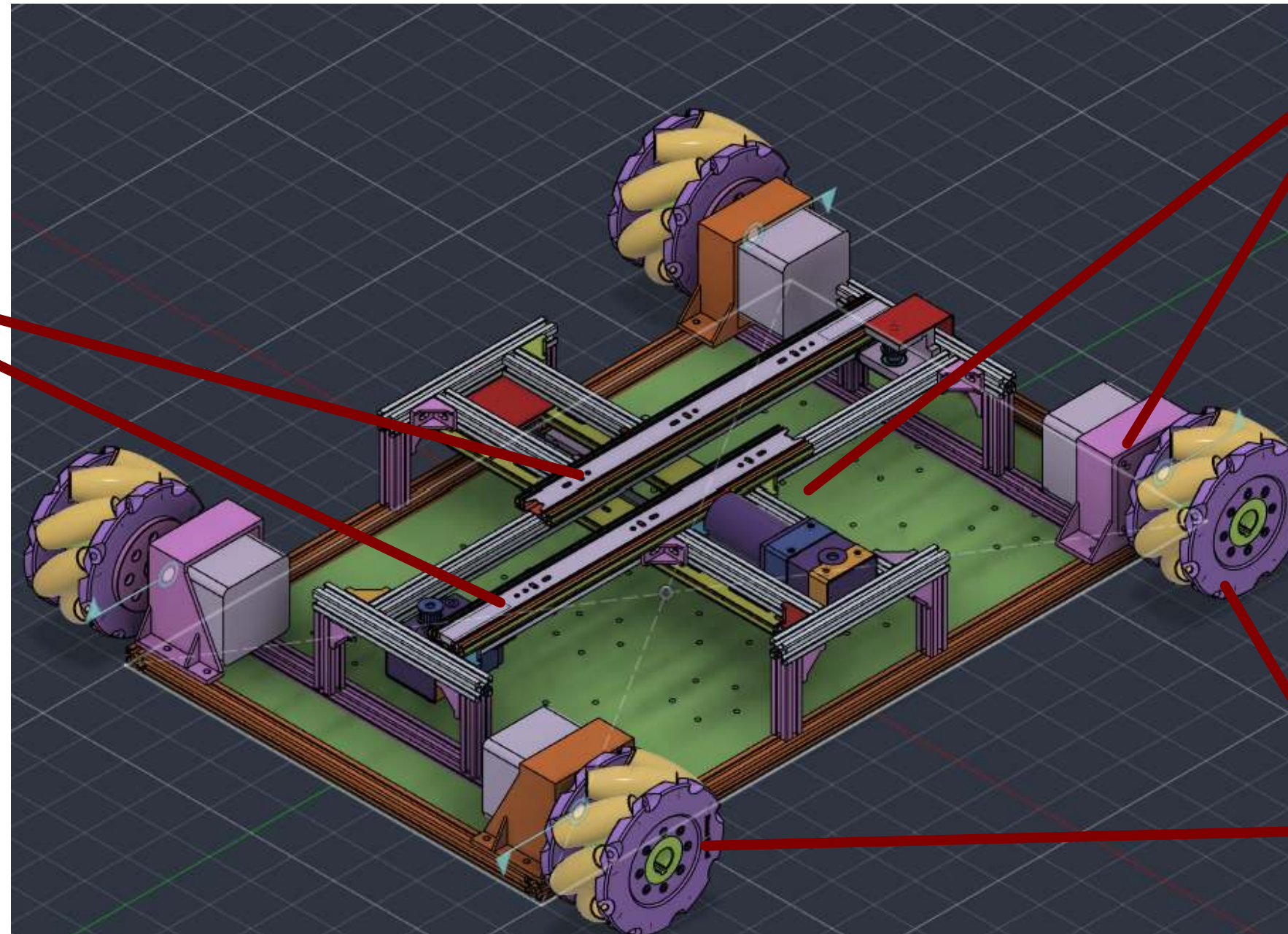
- Firmware & control logic: Claude used to draft and debug the ESP32-S3 state-machine code for arm extension, hook-flip, and caster-alignment control
- System analysis & documentation: Claude assisted with force-balance derivations, unit-checking, and report writing
- Design iteration: AI-assisted review of subsystem trade-offs during concept selection using Gemini and Claude
- Materials selection: Used to compare the function of different motors or switches to find the best for our scenario



CAD Drawings

Arm Extension

Extends and retracts for clamps to reach the sides of the trolley



Overall Material & Structural Aspects of the AMR

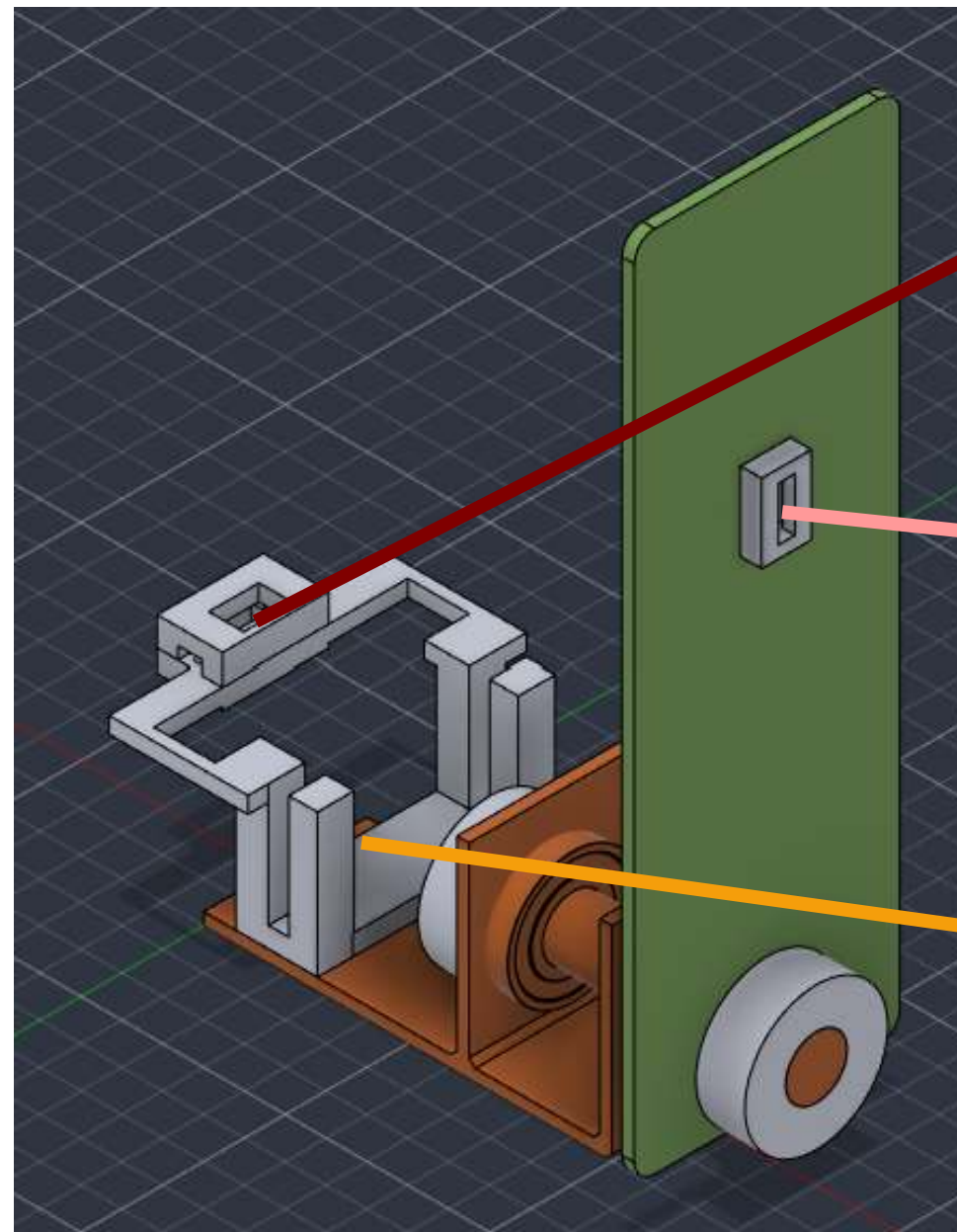
Improved materials for better durability to meet our system requirement

Movement Controls & Power Distribution System

Autonomous adjustment and alignment to be in the centre and underneath the trolley

CAD Drawings

Flipping System



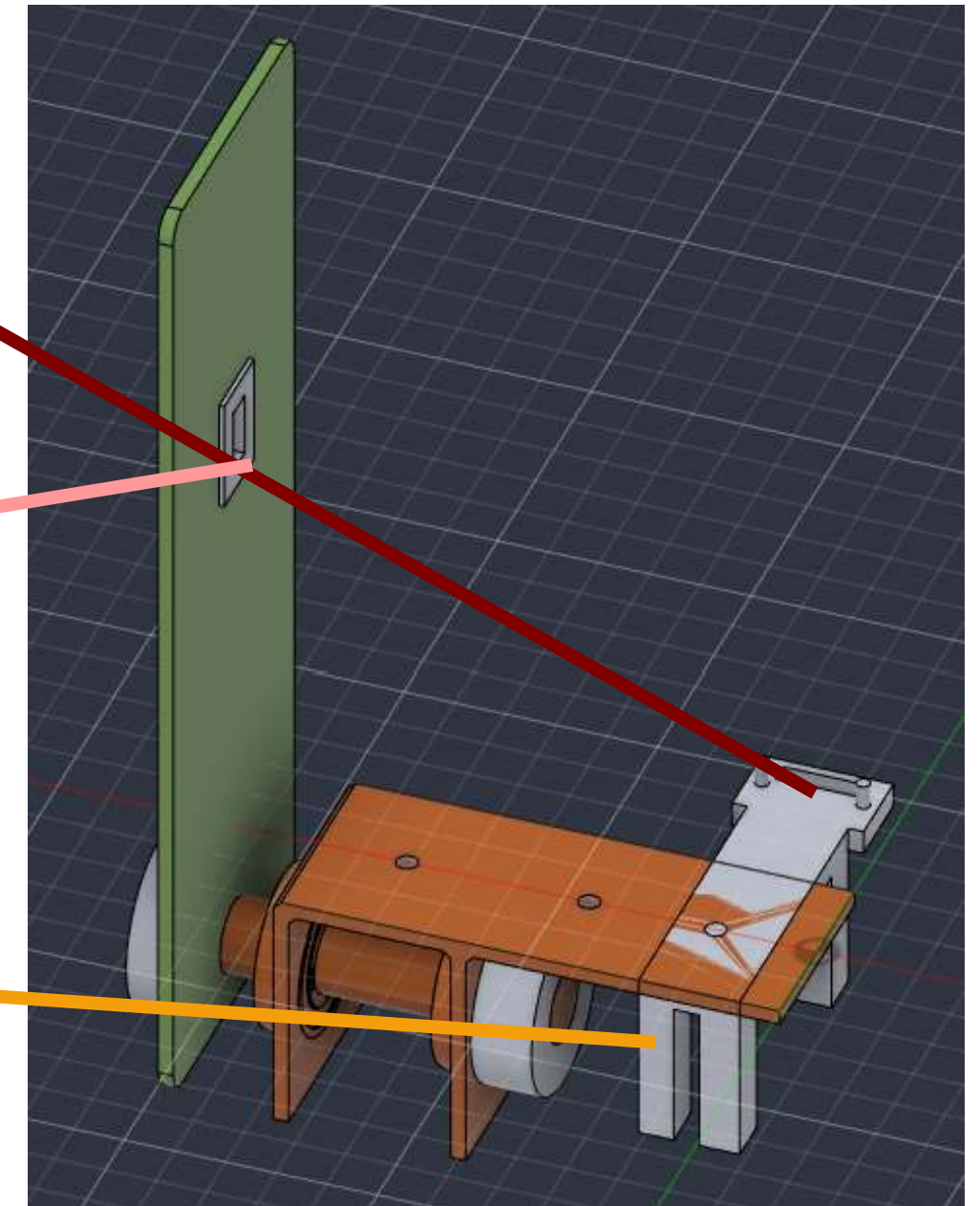
Flipper for the below rails

ToF sensors

Limit Switches

Servo motors

Two different mounting
ways is to minimize the
height of the AMR



Flipper for the above rails

Requirements Summary

Updated functional requirement	Target value	Reasoning	Path to Final Target Assessment
AMR height must fit under trolley clearance	$\leq 200\text{ mm}$	Robot can fit under trolley	Height clearance CAD model validated with metre rule on physical chassis
Clamping arms must withstand holding force	$\geq 300\text{ N}$	Clamp holds load without malfunction/damage	Luggage scale to measure breakaway clamp force under pure horizontal shear.
Clamp vertical reach	$\geq 200\text{ mm}$	Clamp reaches trolley contact/locking point	Physical Measurement of the full clamping system
AMR must move when wheels/castors are misaligned	Works for all 4 castor orientations	Successful movement from worst-case wheel alignment	Manually misalign all 4 castor wheel before moving the of using the AMR to mimic the actual scenario
Able to move	$\geq 100\text{ kg}$	Trolley + Load is at least 100kg	Drive test loaded with 100kg
Time taken for AMR to clamp and dock on trolley	$\leq 1\text{ min}$	Process is quick and efficient enough	Timed docking process

Requirement Changes

Changes of the requirements made since the start of project

Removal of the dust removal system

- Lack of time & money
- Focus effort on the core mobility and docking challenge, which was the higher-impact problem

Fully automated to find the trolley and clamp

- Lack of time
- Lack of money
- Need to apply CV and a lot of sensors for that

Automatic move trolley to the destination

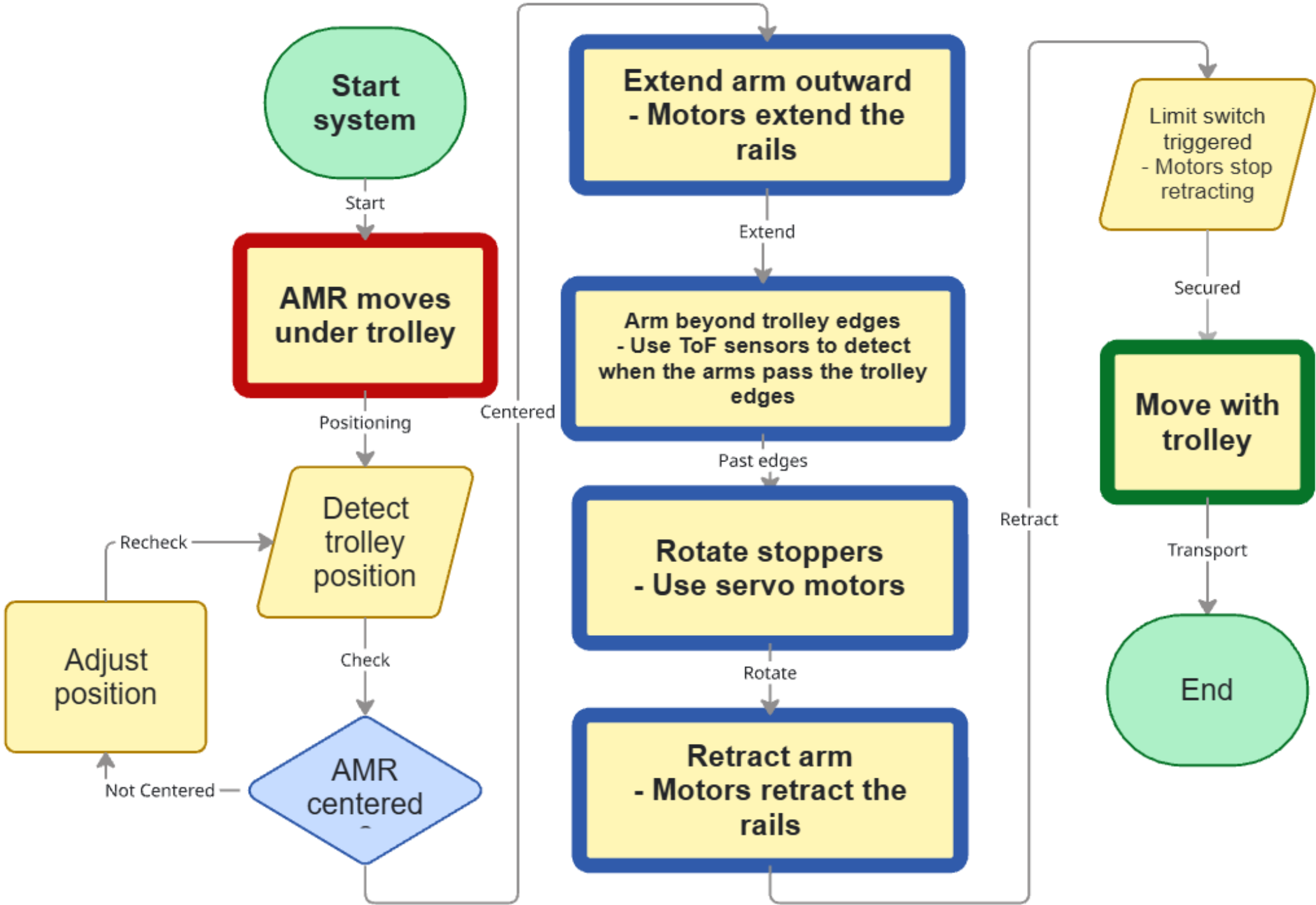
- Need to train the AMR with the map
- Lack of manpower
- Lack of money
- Lack of time

Speed of the trolley when moving

- It's not a must for our AMR

System Analysis Summary

Operational & Control Flow



CRITICAL SUBSYSTEMS & OBJECTIVES

Motor & Gearbox

Predicts: Travel speed ≥ 0.5 m/s at 125 kg

Traction Analysis

Predicts: AMR mass ≥ 25 kg, wheels must not slip under worst-case castor misalignment

Arm Extension & Clamping System

Predicts: Holding force ≥ 300 N

ARM EXTENSION & CLAMPING SYSTEM

Horizontal Pushing (4 Clamps, 1 per side)
Each clamp designed for full 300 N independently
Worst-case horizontal forces

Scenario	Equation	Force
Rolling resistance	$\mu \times m \times g = 0.05 \times 100 \times 9.81$	49 N
Castor misalignment	$\mu \times m \times g = 0.15 \times 100 \times 9.81$	147 N
Emergency braking	$m \times a = 100 \times 2.0$	200 N

Component capacity vs 300 N load

Component	Capacity	Safety Factor
Clamp (3mm Alu, bending)	403N	1.34×
Rail bolts (2× M4 pull-out)	2,500 N	8.3×

Clamp Bending Derivation (Dual 3mm Alu)

Section Modulus (S): $2 \times \frac{70 \times 3^2}{6} = 210mm^3$

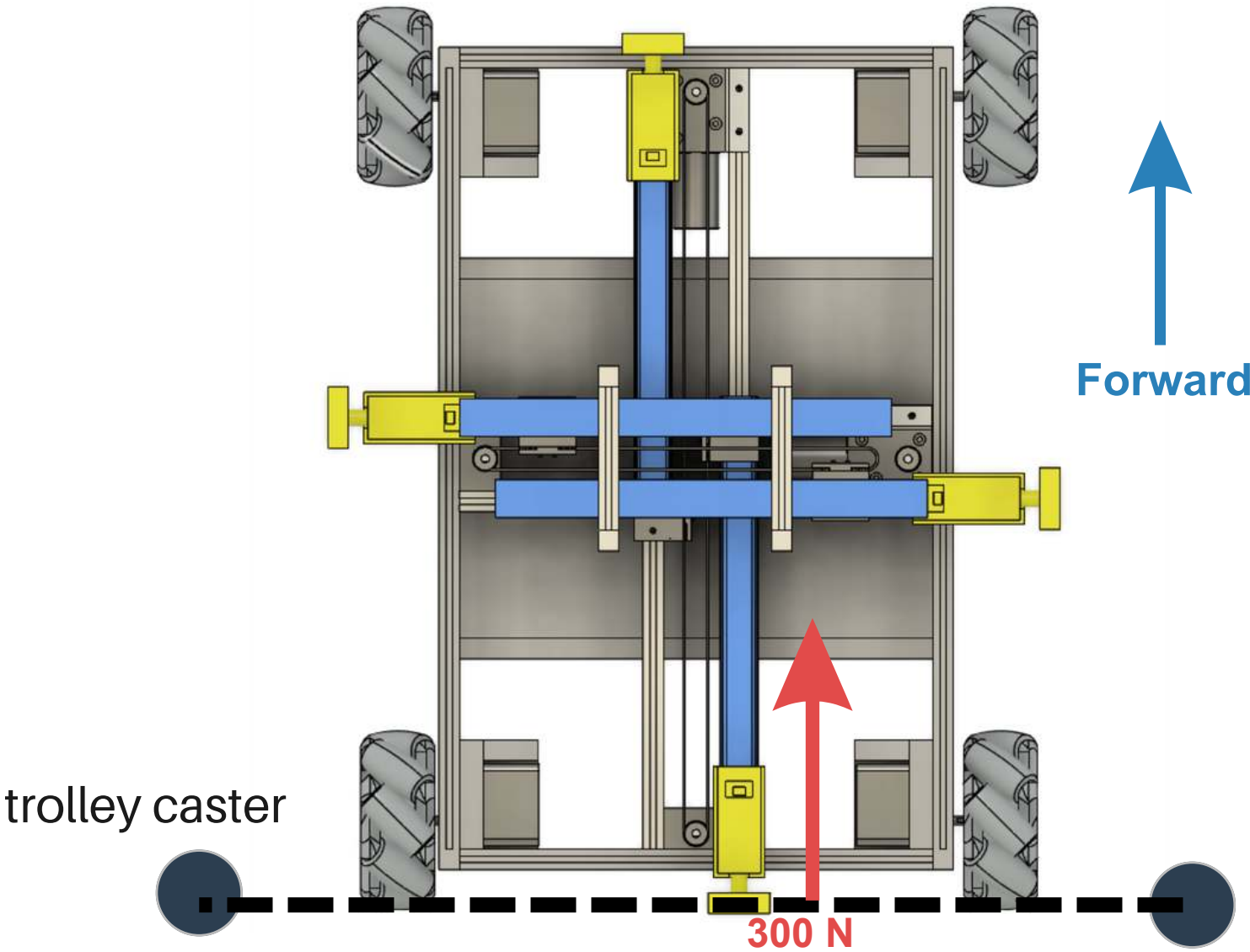
Yield Moment (My): $210 \times 240MPa = 50400Nmm$

Effective Moment Arm (Leff): $50 + \frac{150}{2} = 125mm$

Allowable Force & SF: $F_{cap} = \frac{50400}{125} = 403.2N \Rightarrow SF = 1.34 \times$

✓ Weakest link: Clamps at 1.34× — all components pass

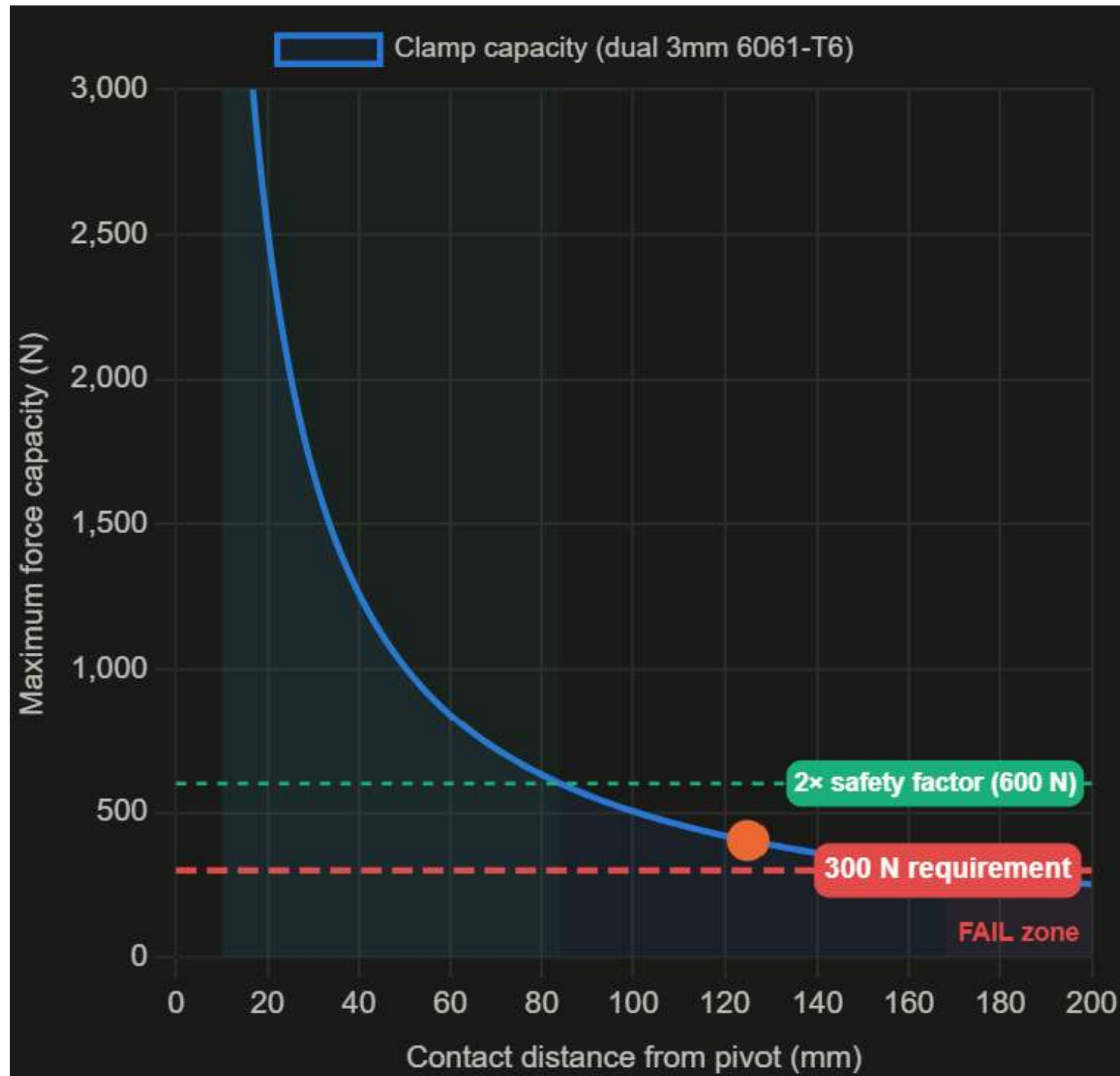
Top-down view — 4 clamp blocks
Worst case: one block takes full 300 N in direction of travel



Assumptions

- μ_{castor} = 0.05 (estimated,)
- $\mu_{\text{misaligned}}$ = 0.15 (estimated, not experimentally validated)
- a_{max} = 2.0 m/s² (traction limit is 1.18 m/s²)
- Rail bolt pull-out = 1,250 N per M4 hammer-head T-nut
- Belt motor holds position during reverse loading

PERFORMANCE PREDICTION (CLAMP BENDING CAPACITY)



- Contact distance depends on trolley frame geometry and varies between trolley models.
- The graph shows clamp capacity across all possible contact points, allowing the design to be validated against any trolley configuration
- At the current design point, $L_{eff} = 125mm$
 - capacity = 403N → safety factor 1.34×
- Clamp passes the 300N requirement but falls below the 2× margin
- Contact beyond 168mm from pivot → clamp fails before reaching 300N
- Weakest link in the system — all other components exceed 8× safety factor

Drive & Layout

Why a mecanum base

- Strafes sideways — lateral centring under the trolley is one direct move, no shuffling
- Rotates in place — zero turning radius; directly answers the tow-tractor “needs turning space” gap from State of the Art
- Same 4 wheels perform approach, orient and centre — no extra actuators

Sensing on the base

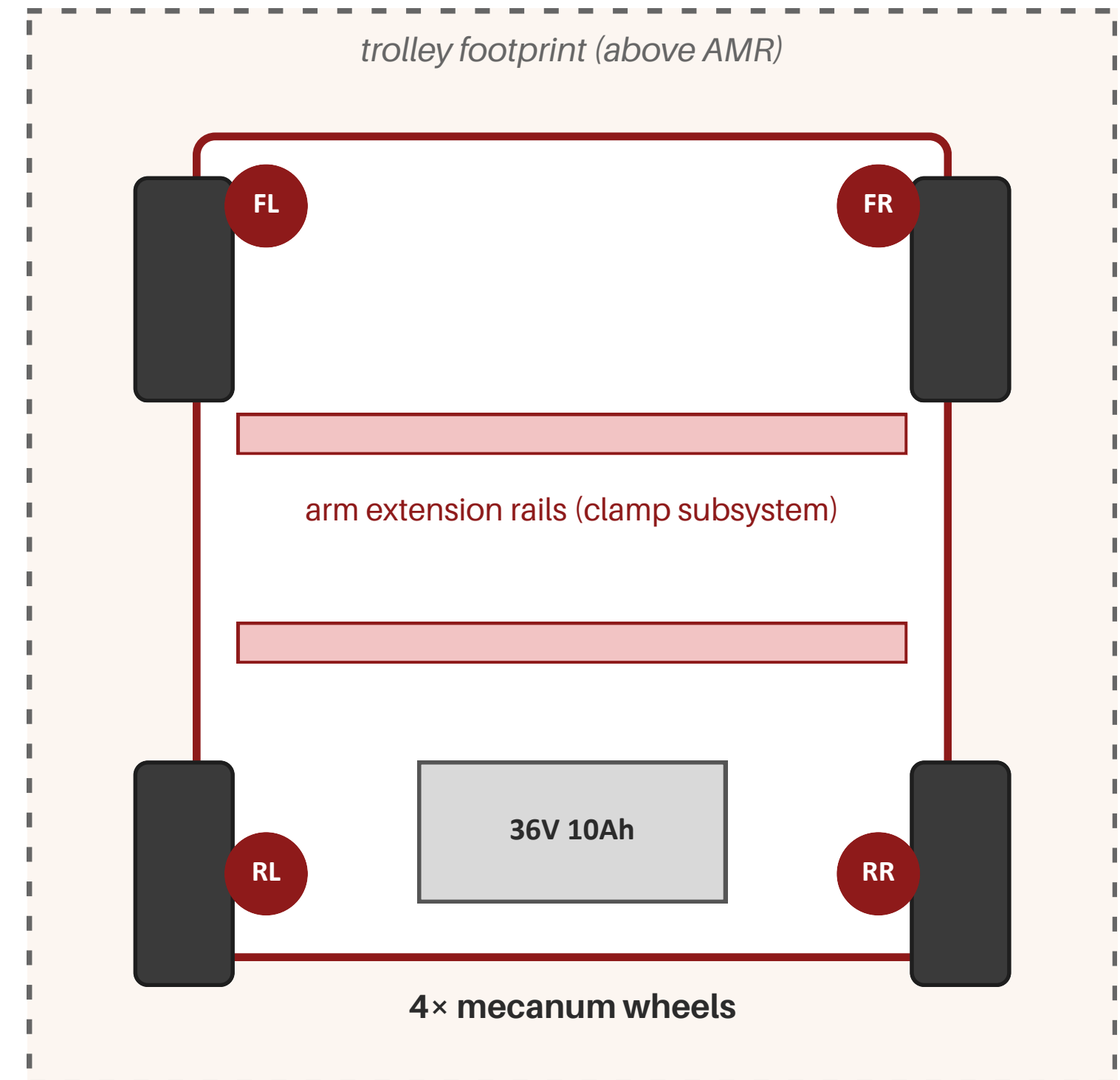
- 4× VL53L0X time-of-flight distance sensors face upward at the corners — front-left, front-right, rear-left, rear-right — connected through a TCA9548A I²C multiplexer. These alone drive the alignment sequence

Power chain

- 36V 10Ah battery feeds three buck converters in parallel — 24V, 12V and 5V rails, each tapped directly off the battery

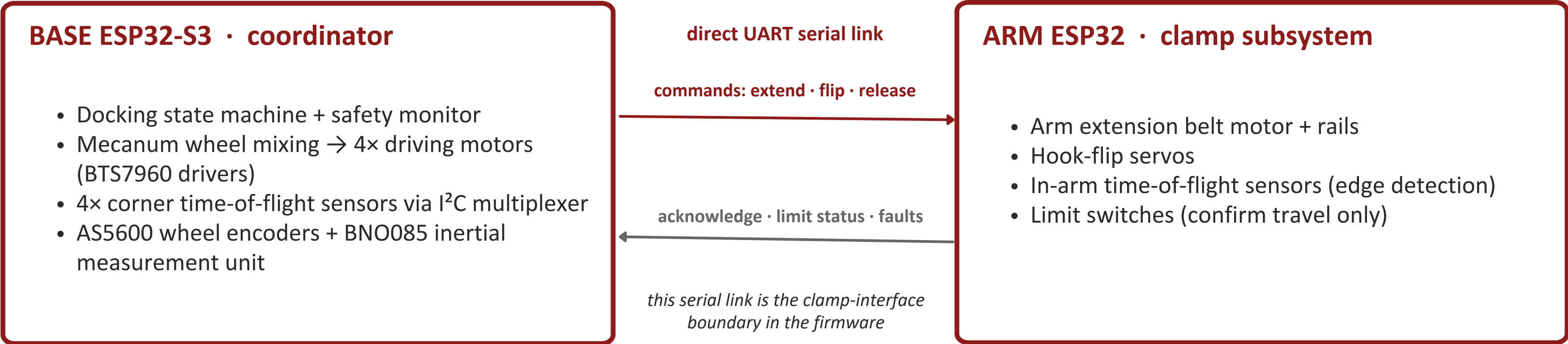
Height budget

- 167 mm measured vs ≤ 200 mm clearance — validated on physical chassis with metre rule



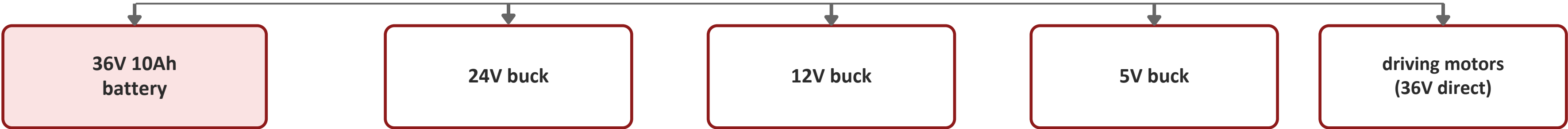
upward-facing VL53L0X time-of-flight distance sensor — one per corner

Dual ESP32 over UART



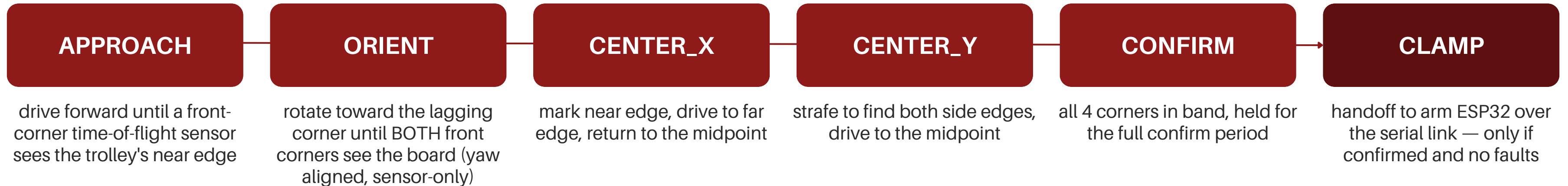
FAILS SAFE: base ESP32 is the single clamp authority — no acknowledgement → no clamp motion · missed response → timeout → fault state (drive braked, clamp stopped)

Power distribution



Driving motors and all three converter rails tap the 36V battery bus directly (parallel, not daisy-chained) — one converter failing cannot drag down another rail.

Autonomous Centring



any fault / per-phase timeout → FAULT: drive disabled + braked, clamp stopped · ABORT clears the latch → IDLE

How a corner “sees” the board

Time-of-flight reading valid AND inside a height band, debounced over several samples — a single spike can't flip it. Solid trolley board → continuous, reliable returns.

Why centring is calibration-proof

Both edges are measured by the same odometry integrator, and we drive to their midpoint — any calibration scale error cancels out.

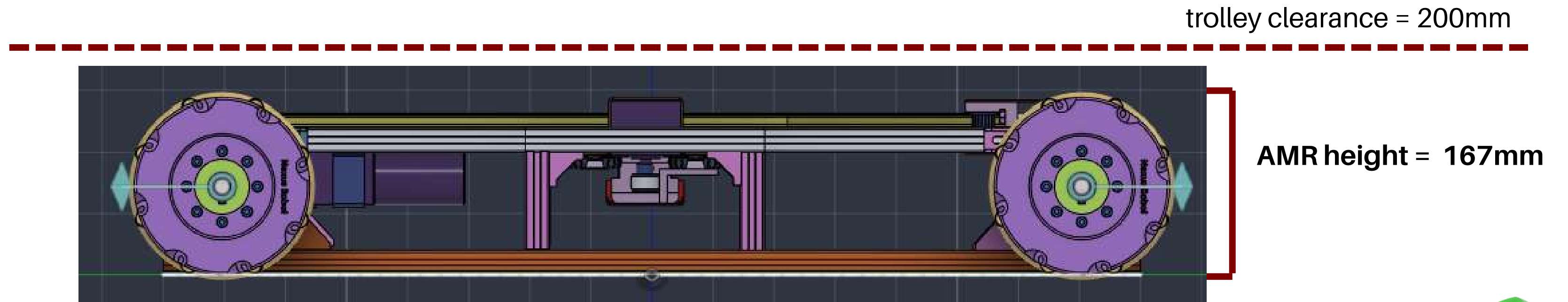
SAFETY INTERLOCK — the clamp physically cannot engage unless:

alignment CONFIRMED (sustained) && no motor / overcurrent fault. Limit switches confirm clamp travel only — they never authorise clamping.

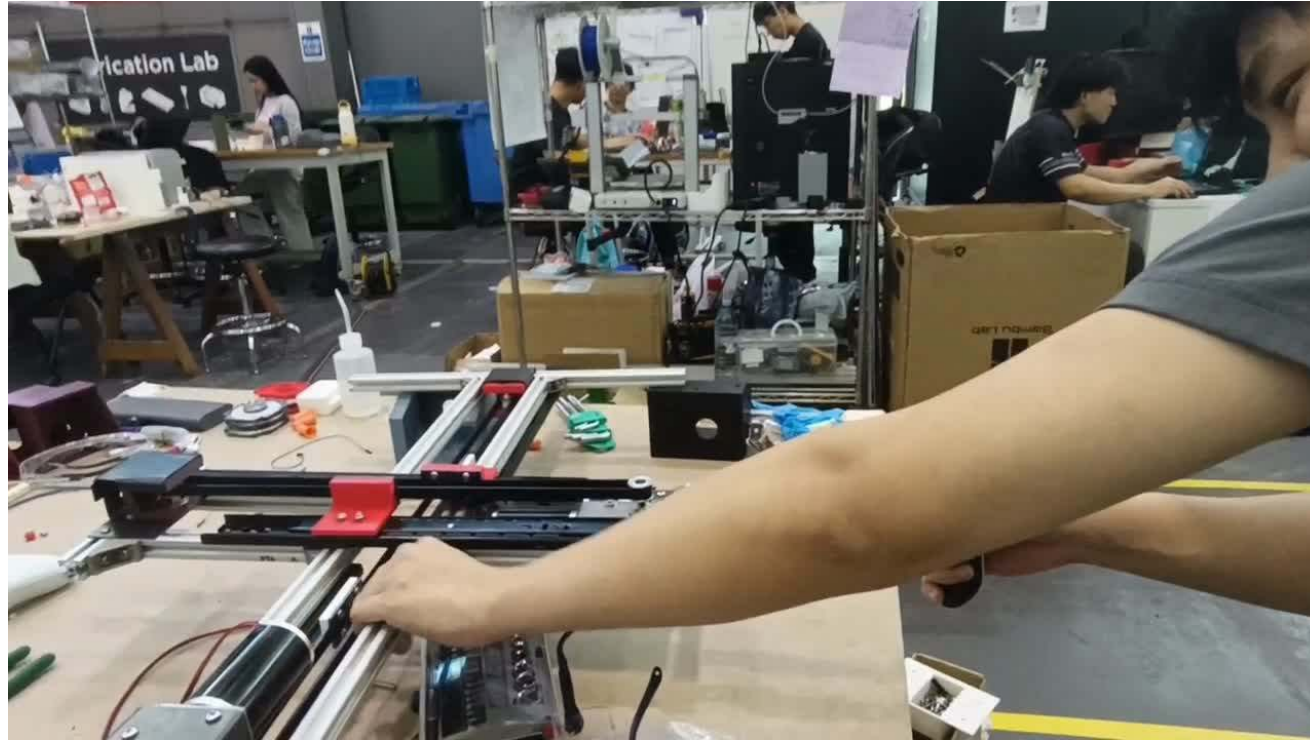
Experiment Results

Height Stack Diagram

- Measurement of our AMR to $\geq 200\text{mm}$ (trolley clearance)



Experiment Results



Experimental Goal & Metric

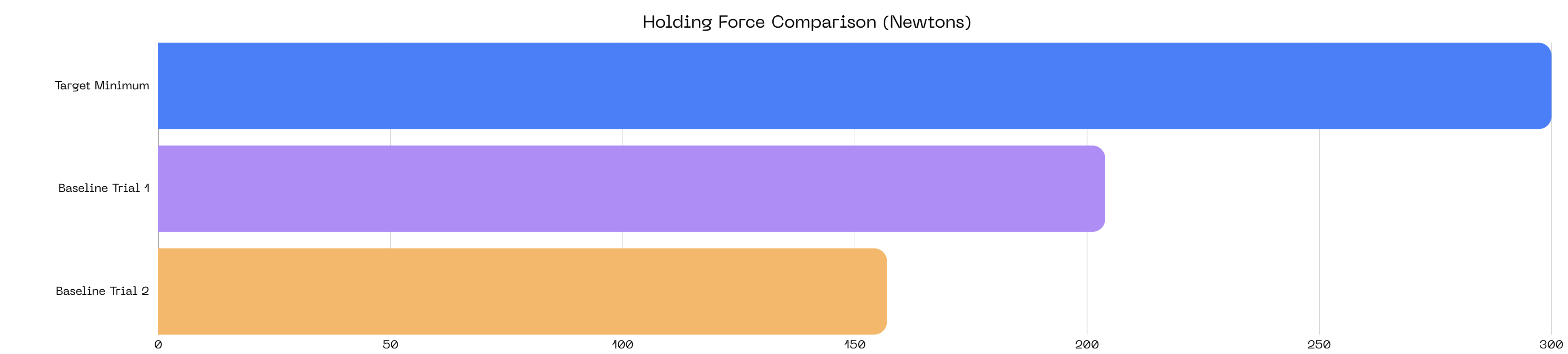
- Verify whether the extension arm joint mechanism satisfies the structural requirement of Holding Force $\geq 300\text{ N}$ prior to rotational slippage:

$$F_{hold} = m \cdot g \geq 300\text{ N} (m_{target} \geq 30.58\text{ kg})$$

TEST PROTOCOL

- **Test Setup Description:** Hand-held tension pull using a calibrated digital luggage scale attached to the extension arm payload point.

Experiment Results



STATISTICAL TRIAL SUMMARY (N = 5 TRIALS PER PHASE)

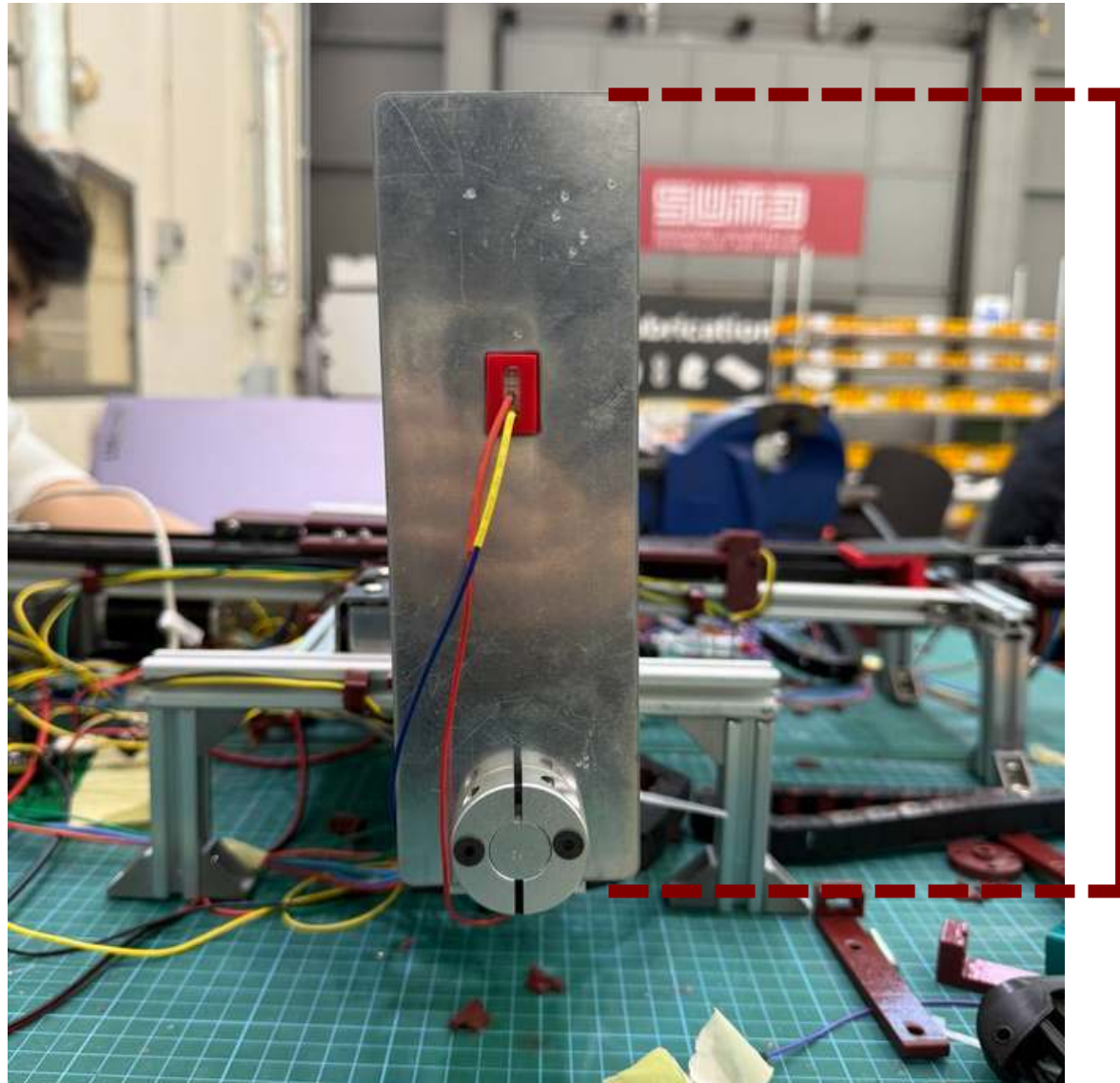
Test Phase	Mean Force	Min Force	Max Force	Std Dev ()	Target Deficit / Surplus	Pass Rate
Baseline Concept	180.4 N (18.4 kg)	156.9 N (16.0 kg)	203.9 N (20.8 kg)	23.5 N	-119.6 N (-39.9%)	0% FAIL (w/o clamp yield)

Observed Failure Modes: Human error in hand-holding introduced off-axis loading and impulse spikes. Failure occurred due to weak anchoring point for the luggage scale. However, the clamping system showed no signs of failure at near 20kg

Experiment Results

Clamp Vertical Reach

- The total height of the clamp with trolley must $\geq 200\text{mm}$



Clamp = 200mm

TOTAL HEIGHT AFTER THE STOPPERS FLIPPED

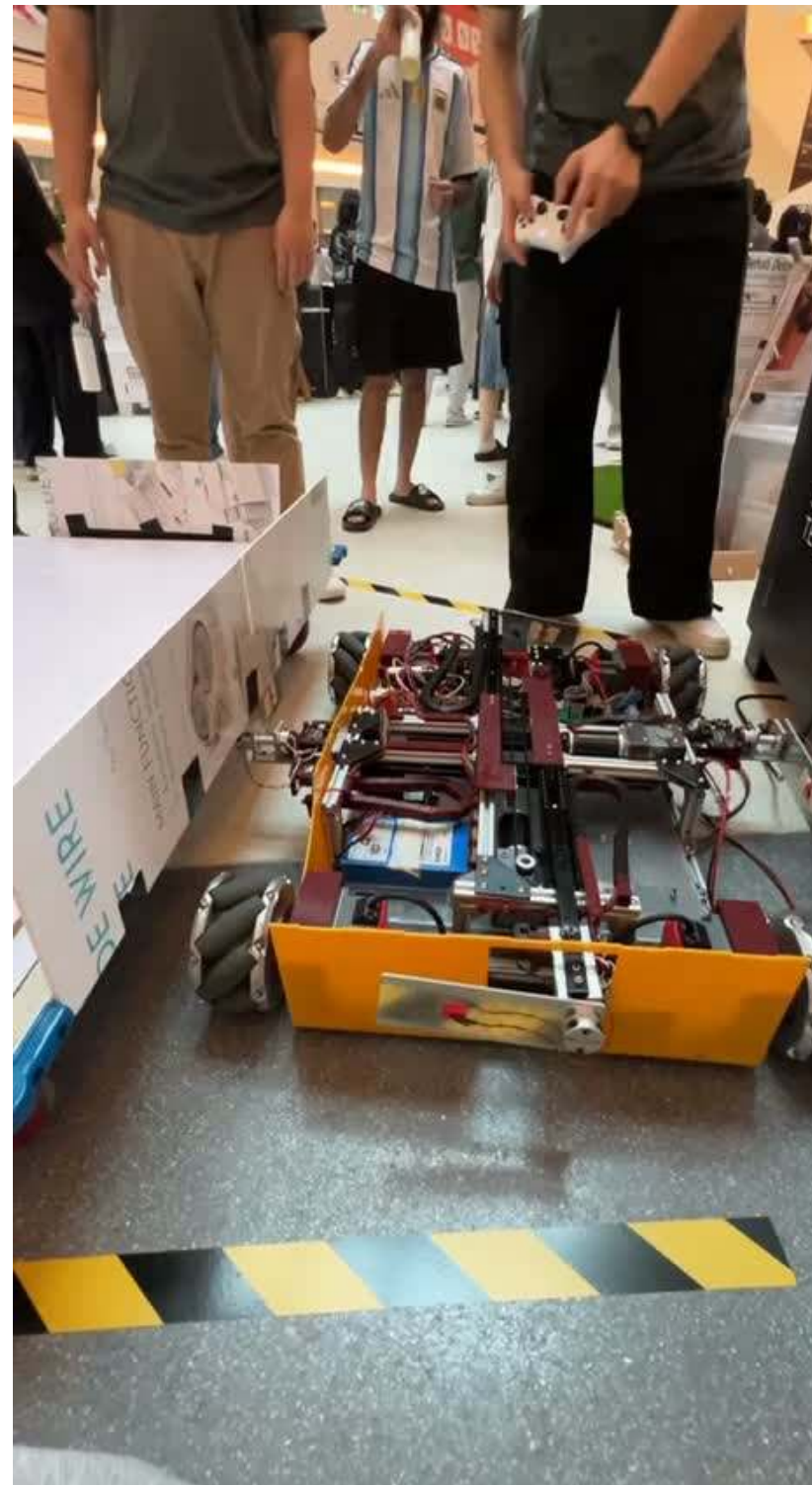
$$200 + 167 = 367\text{mm}$$

High enough to secure the trolley



Experiment Results

Clamping Mechanism



Time Taken for Clamping
Based on testing ≤ 60 s

Within our target time



Experiment Results

Load Testing



Load carry

Load $\geq$ 100kg

Moved with 2 people on top

Can move more than requirement



Experiment Results

**Wheels
misaligned**



Move freely

Works for all 4 castor orientations
Able to move in any directions

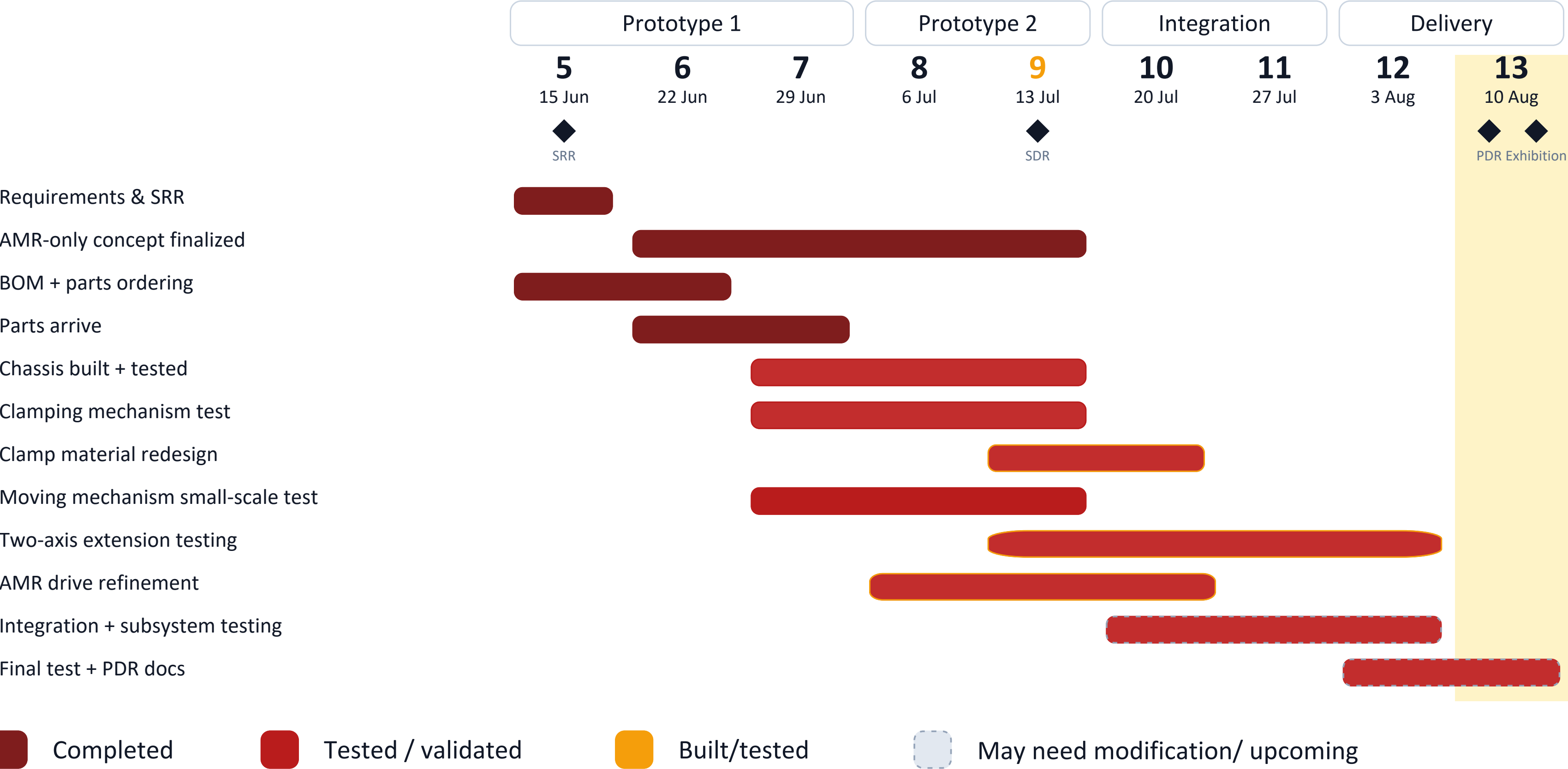
Can move for all 4 castor orientations



Budget Update

Category	Items Included	Estimated Total Cost (SGD)
Mechanical Structure & Clamping System	Turbo Worm Gear Motor, HTD 3M Pulleys & Belts, Drawer Slides, Fasteners (T-Nuts, Brackets, Standoffs, Screws), Aluminum Plates & Extrusions, Bearings, Shaft Collars, Mild Steel Plate, Round Rods	\$652.02
Electrical Wiring & Integration	BLDC Motor, BTS7960 Driver, Solid State Relay, Blade & Glass Fuses with Holders, Terminal Jumpers, Isolated Buck Converter, Brake Resistor, 36V 10Ah Battery + Charger, Wiring Cables, Terminal Blocks, Connectors	\$903.87
Controllers, Sensors & Safety Components	E-Stop Button, Momentary Switch, Limit Switches, Raspberry Pi 5, Time of Flight (ToF) Sensors, ADS1115 ADC, TCA9548A Multiplexer, Resistor Kit, Jumper Wires, Current Sensors, BNO085 IMU Modules, VL53L0X Laser Sensor, Magnetic Encoders	\$415.28
Prototyping & Design Iteration	Claude Subscription, Freight Shipping Fee, JLCPCB Manufacturing, Laser Cutting for Housing, Filament	\$526.19
Total Spent	All paid items based on the purchase record	\$2497.35
Remaining Budget	Budget left after the current and potential purchases	\$2.65

Project schedule - Weeks 5-13



Summary

IMPACT AND INNOVATION

A low-profile AMR that moves existing fabric trolleys from beneath with no trolley fleet redesign and manual docking. This directly cuts manual-push labour and the recurring collision/damage cost on the production floor.

MAIN CONTRIBUTIONS

- Adapts to existing trolleys — no redesign needed
- Autonomous four-sided capture via sensor-driven state machine
- Mechanical stopper carries the load, not the servo

MAJOR ACHIEVEMENTS TO DATE

- ✓ Revised chassis outline & base for a lower, more stable under-trolley fit
- ✓ Arms can extend and lock in place
- ✓ Autonomous auto-align to the centre of the trolley
- ✓ AMR can counter the wheels in different directions

Proposed Future Works

Tracking system

Allow for multiple AMR to work together at the same time without causing any potential incidents

Loaded movement validation

Drive a fully captured, loaded trolley up to 200kg

Mechanical robust

Sturdier arms for repeated testing and higher load. The same applies for the other components making up the AMR

Ability to go even lower than the current height

Able to fit any type of trolley and suitable for any use

Autonomous driving and docking

Autonomous from start to end including moving under the trolley and going from one place to another

Dust filtration system

Target one of the key problems faced by textile companies

Conclusion

Our group has successfully demonstrated that our proposed solution is feasible and useful for our industry partner. Testing and analysis showed that the key subsystems can perform their intended functions.

Further development is needed to make the product fully automated, improve reliability, and validate performance under real operating conditions.

Thank you!

Prototype Showcase / Live Demo

